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United States Patent	12404744
Kind Code	B2
Date of Patent	September 02, 2025
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Wireline retrievable opening prong

Abstract

A subsurface safety valve system may include a tubular valve housing disposable within a wellbore, a primary flow tube moveably secured within the valve housing between an upper position and a lower position, and a secondary valve securable within the valve housing downhole from the primary flow tube and a primary valve. The secondary valve is actuatable between an open position and a closed position. Further, the subsurface safety valve system may include a tubular opening prong securable to the primary flow tube, wherein the opening prong is moveable axially downhole to drive the secondary valve into the open position in response to the primary flow tube moving to the lower position, and wherein the tubular opening prong is movable axially uphole to release the secondary valve to move into the closed position.

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Appl. No.:	18/119214
Filed:	March 08, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240301767 A1	Sep. 12, 2024

Publication Classification

Int. Cl.: E21B34/08 (20060101)

U.S. Cl.:

Field of Classification Search

CPC: E21B (34/08); E21B (2200/05); E21B (34/12); E21B (34/16); E21B (34/106); E21B (34/102)

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Background/Summary

BACKGROUND

(1) After drilling a wellbore in a subterranean formation for recovering hydrocarbons such as oil and gas lying beneath the surface, subsurface safety valves may be placed in the wellbore during completion operations. Subsurface safety valves are generally used as failsafe mechanisms to prevent the uncontrolled release of subsurface production fluids, should a wellbore system experience a loss in containment. Typically, subsurface safety valves are run-in-hole as part of a tubing string. Although several design variations are possible for subsurface safety valves, generally subsurface safety valves comprise flapper-type valves. Unfortunately, flapper-type valves may malfunction and prevent the valve from opening and closing to permit or restrain flow through the valve, respectively, as needed. Further, replacing the subsurface safety valves is costly and time consuming as the valve is generally pulled out-of-hole for repairs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These drawings illustrate certain aspects of the present disclosure and should not be used to limit or define the method.
- (2) FIG. 1 is an elevation view of an example of a well system, in which aspects of the present disclosure may be implemented.
- (3) FIG. 2 is a cross-sectional view of a subsurface safety valve system according to an example configuration having a primary valve and a secondary valve run-in-hole and located at an end sub of the subsurface safety valve system.
- (4) FIG. 3 is a cross-sectional view of the secondary safety valve locked to the end sub.
- (5) FIG. 4 is a cross-sectional view a tubular opening prong and upper key assembly run-in-hole and located in a valve housing of the subsurface safety valve, in accordance with a further example configuration.
- (6) FIG. 5 is a cross-sectional view of the upper key assembly locked to a primary flow tube of the subsurface safety valve.
- (7) FIG. 6 is a cross-sectional view of the primary flow tube driven from an upper position to a lower position to move the tubular opening prong into a position to open the secondary safety valve.
- (8) FIG. 7 is a cross-sectional view of a retrieval tool according to an example configuration, secured to the upper key assembly.
- (9) FIG. 8 is a cross-sectional view of the upper key assembly released and pulled in the uphole direction via the retrieval tool.
- (10) FIG. 9 is a cross-sectional view of the retrieval tool secured to the lower key assembly of the secondary safety valve.
- (11) FIG. 10 is a cross-sectional view of the lower key assembly released and pulled in the uphole direction via the retrieval tool.

DETAILED DESCRIPTION

(12) Disclosed herein are systems and methods for installing, operating, and retrieving a secondary subsurface safety valve. More particularly, example embodiments may include a subsurface safety valve system that includes a primary valve and a secondary valve configured to locate and secure within a valve housing of the subsurface safety valve to restore functionality to the subsurface safety valve in response to malfunctioning or failure of the primary valve. The subsurface safety valve system may include a tubular opening prong for opening and closing the secondary valve. The tubular opening prong may also be secured within the valve housing during operation. However, in the event of failure or malfunction of the secondary valve, both the tubular opening prong and the secondary valve may be retrieved and replaced via a wireline retrieval tool. As such, the secondary valve may be replaced without pulling the subsurface safety valve system out-of-hole, which may reduce costs and downtime in the event of failure or malfunction.

(13) FIG. 1 is an elevation view of a well system, in which aspects of the present disclosure may be implemented. While FIG. 1 generally depicts land-based operations, the principles described herein are equally applicable to subsea operations that employ floating or sea-based platforms and rigs, without departing from the scope of the disclosure. As illustrated, the well system **100** includes a subsurface safety valve system **102** disposed within a wellbore **104** to prevent the uncontrolled release of subsurface production fluids, should the well system **100** experience a loss in containment. Generally, the subsurface safety valve system **102** (e.g., tubing retrievable subsurface safety valve) may be run-in-hole to a desired depth as part of a completion string. Alternatively, the subsurface safety valve system **102** (e.g., wireline retrievable subsurface safety valve) may be lowered to a desired position in the wellbore **104** via wireline.

(14) Moreover, the subsurface safety valve system **102** may be in communication with the surface **106** via a control line **108** (e.g., hydraulic control line, electrical control line, etc.). The control line **108** may provide actuation power to the subsurface safety valve system **102**. As set forth in greater detail below, the actuation power may be provided to the subsurface safety valve system **102**, via the control line **108**, to actuate the subsurface safety valve system **102**. For example, a hydraulic control line may increase a pressure in the control line to ultimately open at least one valve of the subsurface safety valve system **102** to provide a flow path for subsurface production fluids to enter a conduit **110** (e.g., completion tubing, casing, casing liner, etc.). Reducing pressure in the hydraulic control line may ultimately close the at least one valve of the subsurface safety valve system **102** to block the flow path into the conduit **110**.

(15) As set forth above, a primary valve **200** (shown in FIG. 2) of the subsurface safety valve system **102** may fail or malfunction during well operations. In response to failure of the primary valve **200**, a secondary valve (shown in FIG. 2) may be run-in-hole and secured within the subsurface safety valve system **102**. The secondary valve **202** may be lowered into the wellbore **104** via wireline. However, the secondary valve **202** may be lowered into the wellbore **104** via any suitable tool. Further, a tubular opening prong **400** (shown in FIG. 4) for opening and closing the secondary valve **202** may be run-in-hole after the secondary valve **202** is located and secured within the subsurface safety valve system **102**. The tubular opening prong **400** may be run-in-hole via any suitable tool. The tubular opening prong **400** may be considered tubular in that, for example, it may have an outer wall with a generally round cross-section across a portion of its length to conform with an internal diameter (ID) of a structure that it rides in and that it may have an interior that allows for passage of fluid other material therethrough.

(16) FIG. 2 is a cross-sectional view of the subsurface safety valve system **102** according to an example configuration having a primary valve **200** and a secondary valve **202**. For discussion purposes, the subsurface safety valve system **102** is shown run-in-hole and located at an end sub **204** of the subsurface safety valve system **102**. The subsurface safety valve system **102** includes a tubular valve housing **206** disposable within the wellbore **104**. That is, the tubular valve housing **206** may be run-in-hole as part of a completion tubing or lowered into the wellbore **104** via wireline. Moreover, the tubular valve housing **206** may include a main body portion **208**. As illustrated, the tubular valve housing **206** may also include the end sub **204** secured to a downhole end **210** of the main body portion **208**. The end sub **204** may be secured to the main body portion **208** via threaded connection or any other suitable connection.

(17) The subsurface safety valve system **102** also includes a primary flow tube **212**, which may be positioned within the main body portion **208**. In particular, the primary flow tube **212** may be positioned within a flow tube recess **214** formed in an inner surface **216** of the main body portion **208** of the tubular valve housing **206**. As illustrated, the flow tube recess **214**, which receives the primary flow tube **212**, may be formed along a first portion **218** of the inner surface **216** between adjacent portions (e.g., a second portion **220** and a third portion **222**) of the inner surface **216**. The inner surface **216** of the main body portion **208** comprises a variable diameter. In particular, the first portion **218** may have a larger diameter than the adjacent portions (e.g., the second portion **220** and the third portion **222**), such that an upper shoulder **224** is formed at a transition from the second portion **220** to the first portion **218**, and a lower shoulder **226** is formed at a transition from the first portion **218** to the third portion **222** of the inner surface **216**. The flow tube recess **214** may be formed between the upper shoulder **224** and the lower shoulder **226** along the first portion **218** of the inner surface **216**. However, the flow tube recess **214** may be formed in any suitable portion of the main body portion **208** for receiving the primary flow tube **212**.

(18) Further, the flow tube recess **214** may comprise a longer axial length than the primary flow tube **212** such that the primary flow tube **212** is axially moveable within the tubular valve housing **206** between an upper position and a lower position. In the illustrated embodiment, the primary flow tube is disposed between the upper position and the lower position. Moreover, in the

illustrated embodiment, the primary flow tube is disposed between the upper position and the lower position. As illustrated, a diameter of a key body portion **228** of the primary flow tube **212** may include a diameter less than the first portion **218** of the inner surface **216**, but greater than the second portion **220** and third portion **222** of the inner surface **216**. As such, the upper shoulder **224** may contact an upper end **230** of the key body portion **228** to restrain uphole movement of the primary flow tube **212** at the upper position of the primary flow tube **212**. Further, the lower shoulder **226** may contact a lower end **232** of the key body portion **228** to restrain downhole movement of the primary flow tube **212** at the lower position of the primary flow tube **212**. As such, the primary flow tube **212** may be secured between the upper shoulder **224** and the lower shoulder **226** of the main body portion **208**.

(19) Further the primary flow tube **212** may be moveable (e.g., slidable) between the upper shoulder **224** and the lower shoulder **226** in response to actuation power from the control line **108**. As illustrated, the tubular valve housing **206** may further include a piston **234** disposed adjacent the primary flow tube **212**. The piston **234** may be configured to extend at least partially through the upper shoulder **224** to contact the upper end **230** of the key body portion **228** for driving movement of the primary flow tube **212**. The piston **234** may move in response to actuation power in the control line **108** to drive movement of the primary flow tube **212**. For example, the control line (e.g., hydraulic control line) may increase pressure, which may drive the piston **234** in the downhole direction **236**. As the piston **234** moves in the downhole direction, contact between the piston **234** and the upper end **230** of the key body portion **228** may drive the primary flow tube **212** in the downhole direction **236**.

(20) As illustrated, the primary flow tube **212** may include a lower tube portion **238** having a smaller outer diameter than the key body portion **228**. Additionally, the lower tube portion **238** may comprise a smaller diameter than the third portion **222** of the inner surface **216** such that the lower tube portion **238** may move along a central bore **240** of the tubular valve housing **206**. In the illustrated embodiment, the lower tube portion **238** extends downhole from the key body portion **228**. Moreover, as set forth in greater detail below, the lower tube portion **238** may interface with the primary valve **200** to hold the primary valve **200** in an open position. The primary valve **200** is secured within the main body portion **208** of the tubular valve housing **206**. Specifically, the primary valve may be disposed within a valve recess **242** formed in the inner surface **216** of the main body portion **208** of the tubular valve housing **206**, which may be formed downhole from the flow tube recess **214**.

(21) The primary valve **200** may include a primary valve seat **244** secured within the main body portion **208** of the tubular valve housing **206**. Specifically, the primary valve seat **244** may be disposed within the valve recess **242**. Moreover, the primary valve **200** may further include a primary flapper **246** secured to the primary valve seat **244**. The primary flapper **246** is movable between the open position shown here in FIG. 2 and a closed position (e.g., with the primary flapper **246** rotated to engage the primary valve seat **244** and restrain fluid flow through the subsurface safety valve system **102**). Indeed, the lower tube portion **238** of the primary flow tube **212** may contact the primary flapper **246** to drive the primary flapper **246** into the open position, as well as move uphole to release the primary flapper **246** to move into the closed position. However, as set forth above, the primary flapper **246** may fail and/or malfunction during completion operations. To restore functionality to the subsurface safety valve system **102** in response to malfunctioning or failure of the primary valve **200**, a secondary valve **202** may be run-in-hole and secured within the tubular valve housing **206** of the subsurface safety valve system **102**. During installation and operation of the secondary valve **202**, the primary flow tube **212** and/or the tubular opening prong **400** may hold the primary valve **200** in the open position. Accordingly, a permanent lockout tool may not be needed to permanently open the primary valve **200**.

(22) The secondary valve **202** is securable within the tubular valve housing **206** downhole from the primary flow tube **212** and the primary valve **200**. For example, as illustrated, the secondary valve

202 may engage the end sub **204** of the tubular valve housing **206**. The secondary valve **202** may include a lower key assembly **248** having at least one lower key **250**. As set forth in greater detail below, the at least one lower key **250** is expandable into a corresponding lower key slot **252** formed in the tubular valve housing **206** to secure the secondary valve **202** to the tubular valve housing **206**. As illustrated, the lower key slot **252** may be formed in the end sub **204**. Further, the lower key **250** is retractable from the lower key slot **252** to release the secondary valve **202** from the tubular valve housing **206**.

(23) Additionally, the lower key assembly **248** may include a shifting portion **254** and a seal portion **256**. The seal portion **256** may include a mandrel **258** and a seal **260** secured to an outer surface of the mandrel **258**. As illustrated, the seal **260** may interface with the tubular valve housing **206** (e.g., an inner surface of the end sub **204**) to restrain fluid flow between the secondary valve **202** and the inner surface of the tubular valve housing **206**. Moreover, the shifting portion **254** may move axially with respect to the seal portion **256** between an extended position and a mated position. As illustrated, the shifting portion **254** may include a lower anchor portion **262** and a lower support portion **264** rigidly connected to a downhole end of the lower anchor portion **262**. The lower support portion **264** may interface and slide along the mandrel **258** of the seal portion **256**. In the illustrated embodiment, the shifting portion **254** is in the extended position such that the lower support portion **264** is disposed uphole from the at least one lower key **250**, but still connected to the mandrel **258**. The lower key assembly **248** may include a stop for restraining movement of the shifting portion **254** away from the seal portion **256** at the extended position to maintain connection between the shifting portion **254** of and the seal portion **256**.

(24) The shifting portion **254** may include a lower shim **266** extending from a downhole end of the shifting portion **254**. In particular, the lower shim **266** may extend from the downhole end of the lower anchor portion **262**. Further, the lower shim **266** may be secured to the lower support portion **264**. As illustrated, the lower shim **266** may be secured to a radially outer surface of the lower support portion **264**. Alternatively, the lower support portion **264** may comprise the lower shim **266**. As the shifting portion **254** moves from the extended position to the mated position, the lower shim **266** may be driven into contact with the at least one lower key **250**. As set forth in greater detail below, such contact may bias the at least one lower key **250** in a radially outward direction into the corresponding lower key slot **252** formed in the tubular valve housing **206**.

(25) Moreover, the secondary valve **202** may further include a secondary valve seat **268** connected to a downhole end of the lower key assembly **248**. In particular, the secondary valve seat **268** may be connected to a downhole end of the seal portion **256**. Additionally, the secondary valve **202** may include a secondary flapper **270** secured to the secondary valve seat **268**. As set forth in greater detail below, the secondary flapper **270** is actuatable between an open position and a closed position.

(26) FIG. 3 is a cross-sectional view of the secondary valve **202** (e.g., secondary subsurface safety valve) locked to the end sub **204**. As set forth above, the secondary valve **202** is run-in-hole and located within the subsurface safety valve system **102**. In particular, the secondary valve **202** is located such that the at least one lower key **250** of the lower key assembly **248** is axially aligned with the lower key slot **252** formed in the tubular valve housing **206** (e.g., the end sub **204** of the tubular valve housing **206**). Once in position, the shifting portion **254** may be moved from the extended position to the mated position. As the shifting portion **254** moves from the extended position to the mated position, the lower shim **266** may be driven into contact with the at least one lower key **250**. Specifically, a lower end of the lower shim **266** may be wedged between the mandrel **258** of the seal portion **256** and the at least one lower key **250**. As the lower shim **266** continues to be driven in the downhole direction **236** between the mandrel **258** and the at least one lower key **250**, the at least one lower key **250** may be biased in a radially outward direction **300** into the axially aligned lower key slot **252** formed in the tubular valve housing **206**. Further, maintaining the lower shim **266** in position between the at least one lower key **250** and the mandrel

258 may hold the at least one lower key **250** in an expanded position with the at least one lower key **250** secured in the corresponding lower key slot **252**.

(27) Moreover, the secondary valve **202** may further include a lower lock feature **302** to hold the at least one lower key **250** in the expanded position with the at least one lower key **250** secured in the corresponding lower key slot **252**. In particular, the lower lock feature **302** may hold the lower shim **266** in the position between the at least one lower key **250** and the mandrel **258**. The lower lock feature **302** may include a shear pin, a shear screw, and/or any suitable fastener for holding the lower shim **266** in position. The lower lock feature **302** may be disposed in the mandrel **258** and engage the lower shim **266** in response to the lower shim **266** moving into the position between the at least one lower key **250** and the mandrel **258**. Alternatively, the lower lock feature **302** may be disposed in the lower shim **266** and/or the lower support portion **264** and engage the mandrel **258** in response to the lower shim **266** moving into the position between the at least one lower key **250** and the mandrel **258**. The lower lock feature **302** may include a spring (not shown) or any suitable feature for driving the lower lock feature **302** to engage the mandrel **258**, the lower shim **266**, and/or the lower support portion **264**.

(28) FIG. 4 is a cross-sectional view a tubular opening prong **400** and an upper key assembly **402** run-in-hole and located in a tubular valve housing **206** of the subsurface safety valve system **102**, in accordance with a further example configuration. With the secondary valve **202** secured within the subsurface safety valve system **102**, the tubular opening prong **400** may be run-in-hole to the tubular valve housing **206** of the subsurface safety valve system **102**. Generally, the tubular opening prong **400** may interface with the secondary flapper **270** of the secondary valve **202** to open and close the secondary valve **202** during well operations. As set forth in greater detail below, the tubular opening prong **400** may be secured to the primary flow tube **212**, and movement of the primary flow tube **212** may drive the tubular opening prong **400** to contact and drive open the secondary flapper **270**. As such, direct fluid communication, via a fluid communication tool, to the secondary valve **202** may not be necessary to actuate (e.g., to open and close) the secondary valve.

(29) As illustrated, the primary flow tube **212** may have an upper key slot **404** formed in a radially inner surface of the primary flow tube **212**. Specifically, the upper key slot **404** may be formed in a radially inner surface of the key body portion **228** of the primary flow tube **212**. However, the upper key slot **404** may be formed in any suitable portion of the primary flow tube **212**. Further, the subsurface safety valve system **102** may include an upper key assembly **402** having at least one upper key **406**, which is expandable and retractable with respect to a lower body portion **408** of the upper key assembly **402**. As illustrated, the upper key assembly **402** may be lowered into the subsurface safety valve system **102** to a position with the upper key **406** axially aligned with the upper key slot **404**, such that expansion of the upper key **406** may secure the upper key assembly **402** within the upper key slot **404**.

(30) Additionally, the tubular opening prong **400** may be secured to the primary flow tube **212** via the upper key assembly **402**. The tubular opening prong **400** may be secured to the upper key assembly **402** via a threaded connection and/or any suitable fastener. Further, the tubular opening prong **400** may be secured to a downhole end of the upper key assembly **402**. Alternatively, the tubular opening prong **400** may be secured to the primary flow tube **212** directly or via any suitable fastener such that movement of the primary flow tube **212** may drive the tubular opening prong **400** to move to contact and drive open the secondary flapper **270** of the secondary valve.

(31) Moreover, the upper key assembly may have an upper body portion **410** and the lower body portion **408**. The upper body portion **410** may include an upper anchor portion **412** and an upper support portion **414** rigidly connected to a downhole end of the upper anchor portion **412**. The upper body portion **410** may move axially with respect to the lower body portion **408** between an extended position and a collapsed position. Specifically, the upper support portion **414** may move with respect to the lower body portion **408** of the upper key assembly **402** between an extended position and a collapsed position. In the illustrated embodiment, the upper body portion **410** is

disposed in the extended position such that the upper support portion **414** is disposed uphole from the at least one upper key **406** but is still connected to the lower body portion **408**. The upper key assembly **402** may include a stop for restraining movement of the upper body portion **410** away from the lower body portion **408** at the extended position to maintain connection between the upper body portion **410** of and the lower body portion **408**.

(32) The upper body portion **410** may further include an upper shim **416** extending from a downhole end of the upper body portion **410**. In particular, the upper shim **416** may extend from the downhole end of the upper anchor portion **412**. Further, the upper shim **416** may be secured to the upper support portion **414**. As illustrated, the upper shim **416** may be secured to a radially outer surface of the upper support portion **414**. Alternatively, the upper support portion **414** may comprise the upper shim **416**. As the upper body portion **410** moves from the extended position to the mated position (shown in FIG. 5), the upper shim **416** may be driven into contact with the at least one upper key **406**. As set forth in greater detail below, such contact may bias the at least one upper key **406** in a radially outward direction into the corresponding upper key slot **404** formed in the primary flow tube **212**.

(33) FIG. 5 is a cross-sectional view of the upper key assembly **402** locked to a primary flow tube **212** of the subsurface safety valve system **102**. As set forth above, the tubular opening prong **400** is secured to the upper key assembly **402** and the upper key assembly **402** is lowered into the subsurface safety valve system **102** to a position with the upper key **406** axially aligned with the upper key slot **404** formed in the primary flow tube **212**. Once in position, the upper body portion **410** may be moved from the extended position to the collapsed position. As the upper body portion **410** moves from the extended position to the collapsed position, the upper shim **416** may be driven into contact with the at least one upper key **406**. Specifically, a lower end of the upper shim **416** may be wedged between the lower body portion **408** and the at least one upper key **406**. As the upper shim **416** continues to be driven in the downhole direction **236** between the lower body portion **408** and the at least one upper key **406**, the at least one upper key **406** may be biased in the radially outward direction into the axially aligned upper key slot **404** formed in the primary flow tube **212**. Further, maintaining the upper shim **416** in position between the at least one upper key **406** and the lower body portion **408** may hold the at least one upper key **406** in the expanded position with the at least one upper key **406** secured in the corresponding upper key slot **404**.

(34) Moreover, the upper key assembly **402** may further include an upper lock feature **500** to hold the at least one upper key **406** in the expanded position with the at least one upper key **406** secured in the corresponding upper key slot **404**. In particular, the upper lock feature **500** may hold the upper shim **416** in the position between the at least one upper key **406** and the lower body portion **408**. The upper lock feature **500** may include a pin, a shear screw, and/or any suitable fastener for holding the upper shim **416** in position. The upper lock feature **500** may be disposed in the lower body portion **408** and engage the upper shim **416** in response to the upper shim **416** moving into the position between the at least one upper key **406** and the lower body portion **408**. Alternatively, the upper lock feature **500** may be disposed in the upper shim **416** and/or the upper support portion **414** and engage the lower body portion **408** in response to the upper shim **416** moving into the position between the at least one upper key **406** and the lower body portion **408**. The upper lock feature **500** may include a spring (not shown) or any suitable feature for driving the upper lock feature **500** to engage the lower body portion **408**, the upper shim **416**, and/or the upper support portion **414**.

(35) FIG. 6 is a cross-sectional view of the primary flow tube **212** driven from an upper position to a lower position to move the tubular opening prong **400** into a position to open the secondary valve **202**. As set forth above, the subsurface safety valve system **102** may include a piston **234** disposed within the tubular valve housing **206**. The piston **234** is moveable between a first and second position in response to actuation power (e.g., a threshold fluid pressure in the control line **108**). Further, movement of the piston **234** from the first position to the second position may drive the

primary flow tube **212** from the upper position to the lower position. That is, the piston **234** may interface with the primary flow tube **212** to drive the primary flow tube **212** in the downhole direction **236**. Alternatively, the piston **234** may be magnetically connected to the primary flow tube **212** to drive movement of the primary flow tube **212**. Further, additional or alternative features may be incorporated such that movement of the piston **234** may drive movement of the primary flow tube **212**. Moreover, as set forth above, the tubular opening prong **400** is secured to the primary flow tube **212** via the upper key assembly **402**. As such, moving the primary flow tube **212** to the lower position may drive the tubular opening prong **400** axially downhole into the secondary valve **202** and into the secondary flapper **270**, which may drive the secondary flapper **270** into an open position.

(36) During well operations, actuation power may be released (e.g., fluid pressure in the control line **108** may be lowered below the threshold pressure) to move the primary flow tube **212** to the upper position, which may release the secondary flapper **270** to move to the closed position. In particular, the subsurface safety valve system **102** may include a spring assembly **600** disposed in the tubular valve housing **206**. The spring assembly **600** may bias the piston **234** with sufficient force to move the piston **234** from the second position toward the first position in response to the fluid pressure in the control line **108** being below the threshold pressure. As such, the primary flow tube **212** may be biased in the uphole direction **602** toward the upper position in the absence of sufficient pressure in the control line **108**, which may close the secondary flapper **270** of the secondary valve **202**. Alternatively, as set forth above, the secondary valve **202** may be closed in response to releasing the upper key assembly **402** and pulling the upper key assembly **402** and the tubular opening prong **400** in the uphole direction **602** with respect to the secondary valve **202**.

(37) FIG. 7 is a cross-sectional view of a retrieval tool **700** according to an example configuration, secured to the upper key assembly **402**. As set forth above, in the event of malfunction or failure of the secondary valve **202** of the subsurface safety valve system **102**, the secondary valve **202**, the tubular opening prong **400**, and the upper key assembly **402**, may be removed and replaced without pulling the tubular valve housing **206** of the subsurface safety valve system **102** out of hole. As illustrated, the upper body portion **410** of the upper key assembly **402** may comprise the upper anchor portion **412** having an upper lip **702** formed in a radially inner surface of the upper body portion **410**. In particular, the upper lip **702** may be formed in the radially inner surface of the upper anchor portion **412** of the upper body portion **410**. The upper lip **702** may be formed via a variable inner diameter of the radially inner surface of the upper anchor portion **412**. That is, an upper end **704** of the radially inner surface of the upper anchor portion **412** may have a greater diameter than a lower end **706** of the radially inner surface of the upper anchor portion **412**, which may form the upper lip **702** at the transition between the upper end **704** and the lower end **706** of the radially inner surface of the upper anchor portion **412**. Moreover, the upper lip **702** may interface with a retrieval tool **700** (e.g., wireline retrieval tool).

(38) As illustrated, the retrieval tool **700** may be lowered into the upper key assembly **402** and interface with the upper lip **702**. The retrieval tool **700** may jar the upper key assembly **402** by driving the retrieval tool **700** in the uphole direction **602** while interfaced with the upper lip **702** such that driving force applied to the retrieval tool **700** is transferred to the upper key assembly **402** via the interface. Such force on the upper key assembly **402** may shear the upper lock feature **500** holding the upper shim **416** in position. With the upper lock feature **500** sheared, continual upward force on the upper lip **702** via the retrieval tool **700**, may pull the upper body portion **410** of the upper key assembly **402** from the collapsed position to the extended position. As set forth above, moving the upper body portion **410** to the extended position may pull the upper shim **416** uphole from the upper key **406** such that the upper key **406** may radially retract toward the lower body portion **408** of the upper key assembly **402** from the upper key slot **404**; thereby, releasing the upper key assembly **402** from the primary flow tube **212**.

(39) FIG. 8 is a cross-sectional view of the upper key assembly **402** released and pulled in the

uphole direction **602** via the retrieval tool **700**. As illustrated, the upper key assembly **402** of the subsurface safety valve system **102** is in the extended position and released from the upper key slot **404** formed in the primary flow tube **212**. With the upper key assembly **402** released (e.g., with the upper key **406** retracted from the upper key slot **404**), the retrieval tool **700** may pull the upper key assembly **402** and the tubular opening prong **400** out of hole. Moreover, as illustrated, moving the tubular opening prong **400** axially uphole (e.g., out of hole) may release the secondary valve **202** to move into the closed position as illustrated.

(40) FIG. **9** is a cross-sectional view of the retrieval tool **700** secured to the lower key assembly **248** of the secondary valve **202**. As set forth above, in the event of malfunction or failure of the secondary valve **202**, the secondary valve **202**, the tubular opening prong (shown in FIG. **8**), and the upper key assembly **402** (shown in FIG. **8**), may be removed and replaced without pulling the tubular valve housing **206** of the subsurface safety valve system **102** out of hole. As illustrated, the tubular opening prong **400** and the upper key assembly **402** have been pulled out of hole and the retrieval tool **700** is lowered into the secondary valve **202** to retrieve the secondary valve **202**.

(41) As illustrated, the shifting portion **254** of the lower key assembly **248** may comprise the lower anchor portion **262** having an inner lip **900** formed in a radially inner surface of the shifting portion **254**. In particular, the inner lip **900** may be formed in the radially inner surface of the lower anchor portion **262** of the shifting portion **254**. The inner lip **900** may be formed via a variable inner diameter of the radially inner surface of the lower anchor portion **262**. That is, an upper end **902** of the radially inner surface of the lower anchor portion **262** may have a greater diameter than a lower end **904** of the radially inner surface of the lower anchor portion **262**, which may form the inner lip **900** at the transition between the upper end **902** and the lower end **904** of the radially inner surface of the lower anchor portion **262**. Moreover, the inner lip **900** may interface with the retrieval tool **700** (e.g., wireline retrieval tool).

(42) As illustrated, the retrieval tool **700** may be lowered into the lower key assembly **248** of the secondary valve **202** and interface with the inner lip **900**. The retrieval tool **700** may jar the lower key assembly **248** by driving the retrieval tool **700** in the uphole direction **602** while interfaced with the inner lip **900** such that driving force applied to the retrieval tool **700** is transferred to the lower key assembly **248** via the interface. Such forces on the lower key assembly **248** may shear the lower lock feature **302** holding the lower shim **266** in position. With the lower lock feature **302** sheared, continual upward force on the inner lip **900** via the retrieval tool **700**, may pull the shifting portion **254** of the lower key assembly **248** from the mated position to the extended position. As set forth above, moving the shifting portion **254** to the extended position may pull the lower shim **266** uphole from the lower key **250** such that the lower key **250** may radially retract toward the seal portion **256** of the lower key assembly **248** from the lower key slot **252**; thereby, releasing the lower key assembly **248** and the secondary valve **202** from the tubular valve housing **206** (e.g., the end sub **204**).

(43) FIG. **10** is a cross-sectional view of the lower key assembly **248** released and pulled in the uphole direction **602** via the retrieval tool **700**. As illustrated, the lower key assembly **248** is in the extended position and released from the lower key slot **252** formed in the tubular valve housing **206** (e.g., the end sub **204**). The lower key assembly **248** and the secondary valve **202** may have a smaller outer diameter than the inner surface of the primary flow tube **212** and an inner surface of the tubular valve housing **206**. Accordingly, with the lower key assembly **248** released (e.g., with the lower key **250** retracted from the lower key slot **252**), the retrieval tool **700** may pull the lower key assembly **248** and the secondary valve **202** uphole through the subsurface safety valve system **102** and out of hole. Once removed, an alternative secondary valve may be lowered into the subsurface safety valve system **102** to replace the removed secondary valve **202**.

(44) Accordingly, the present disclosure may provide a subsurface safety valve system having a retrievable secondary valve, as well as an upper key assembly and tubular opening prong for operating the secondary valve. The systems and methods may include any of the various features

disclosed herein, including one or more of the following statements. Statement 1. A subsurface safety valve system, comprising: a tubular valve housing disposable within a wellbore; a primary flow tube moveably secured within the valve housing between an upper position and a lower position; a secondary valve securable within the valve housing downhole from the primary flow tube and a primary valve, wherein the secondary valve is actuatable between an open position and a closed position; and a tubular opening prong securable to the primary flow tube, wherein the tubular opening prong is moveable axially downhole to drive the secondary valve into the open position in response to the primary flow tube moving to the lower position, and wherein the tubular opening prong is movable axially uphole to release the secondary valve to move into the closed position. Statement 2. The subsurface safety valve system of statement 1, wherein the tubular valve housing comprises a main body portion and an end sub secured to a downhole end of the main body portion, and wherein the secondary valve is engageable the end sub to secure the secondary valve to the end sub of the tubular valve housing. Statement 3. The subsurface safety valve system of any preceding statement, wherein the primary valve is secured within the main body portion of the tubular valve housing. Statement 4. The subsurface safety valve system of any preceding statement, wherein the primary valve comprises: a primary valve seat secured within a main body portion of the valve housing; and a primary flapper secured to the primary valve seat, wherein the tubular opening prong is configured to hold the primary flapper in an open position. Statement 5. The subsurface safety valve system of any preceding statement, further comprising a fluid control line and piston disposed within the tubular valve housing, wherein the piston is moveable between a first and second position in response to fluid pressure in the fluid control line, and wherein movement of the piston from the first position to the second position drives the primary flow tube from the upper position to the lower position. Statement 6. The subsurface safety valve system of any preceding statement, further comprising a spring assembly disposed in the valve housing, wherein the spring assembly is configured to bias the piston from the second position toward the first position. Statement 7. The subsurface safety valve system of any preceding statement, further comprising an upper key assembly having at least one upper key, wherein the at least one upper key is expandable into a corresponding upper key slot formed in the primary flow tube to secure the upper key assembly to the primary flow tube, wherein the upper key is retractable from the upper key slot to release the upper key assembly from the primary flow tube, and wherein the tubular opening prong is secured to a lower end of the upper key assembly. Statement 8. The subsurface safety valve system of any preceding statement, wherein the upper key assembly comprises an upper body portion and a lower body portion, wherein the upper body portion is axially moveable with respect to the lower body portion between an extended position and a collapsed position. Statement 9. The subsurface safety valve system of any preceding statement, wherein the upper body portion comprises a shim extending from a downhole end of the upper body portion, wherein the shim contacts the at least one upper key in the collapsed position to bias the at least one upper key in a radially outward direction into the corresponding upper key slot formed in the primary flow tube. Statement 10. The subsurface safety valve system of any preceding statement, further comprising an upper lock feature configured to engage and hold the shim in the collapsed position, wherein the upper lock feature comprises a shear pin, a shear screw, or some combination thereof. Statement 11. The subsurface safety valve system of any preceding statement, wherein the upper body portion comprises a hollow mandrel having an upper lip formed in a radially inner surface of the upper body portion, wherein the upper lip is configured to interface with a retrieval tool, and wherein the retrieval tool is configured to jar the upper body portion, via the upper lip, to shear an upper lock feature holding the upper body portion in the collapsed position. Statement 12. The subsurface safety valve system of any preceding statement, wherein the secondary valve comprises: a lower key assembly having at least one lower key, wherein the at least one lower key is expandable into a corresponding lower key slot formed in the valve housing to secure the secondary valve to the valve housing, and wherein the lower key is retractable from the lower key

slot to release the secondary valve from the valve housing; a secondary valve seat connected to a downhole end of the lower key assembly; and a secondary flapper secured to the secondary valve seat, wherein the secondary flapper is actuatable between an open position and a closed position.

Statement 13. The subsurface safety valve system of any preceding statement, wherein the lower key assembly comprises a shifting portion and a seal portion, wherein the seal portion comprises a mandrel and a seal secured to an outer surface of the mandrel, wherein the seal is configured to interface with the valve housing, and wherein the shifting portion is moveable axially with respect to the seal portion between an extended position and a mated position.

Statement 14. The subsurface safety valve system of any preceding statement, wherein the shifting portion comprises a shim extending from a downhole end of the shifting portion, wherein the shim is configured to contact the at least one lower key in the mated position to bias the at least one lower key in a radially outward direction into the corresponding lower key slot formed in the valve housing.

Statement 15. The subsurface safety valve system of any preceding statement, wherein the shifting portion comprises a hollow mandrel having an inner lip formed at an uphole end of the shifting portion, wherein the inner lip is configured to interface with a retrieval tool with the tubular opening prong removed from the wellbore, and wherein the retrieval tool is configured to jar the shifting portion, via the inner lip, to shear a lower lock feature holding the shifting portion in the mated position.

Statement 16. A subsurface safety valve system, comprising: a tubular valve housing disposable within a wellbore, the valve housing having a lower key slot; a primary flow tube axially moveable within the valve housing between an upper position and a lower position, the primary flow tube having an upper key slot formed in a radially inner surface of the primary flow tube; a secondary valve securable within the valve housing downhole from the primary flow tube and a primary valve, wherein the secondary valve comprises a lower key assembly having a lower key configured to expand into the corresponding lower key slot formed in the valve housing to secure the secondary valve to the valve housing; a secondary valve seat connected to a downhole end of the lower key assembly; a secondary flapper secured to the secondary valve seat, wherein the secondary flapper is actuatable between an open position and a closed position; an upper key assembly comprising an upper key, wherein the upper key is expandable into the upper key slot to secure the upper key assembly to the primary flow tube; and a tubular opening prong secured to a downhole end of the upper key assembly, wherein the opening prong is moveable axially downhole to drive the secondary flapper into the open position in response to the primary flow tube moving to the lower position, and wherein the opening prong is moveable axially uphole to release the secondary flapper to move into the closed position.

Statement 17. The subsurface safety valve system of statement 16, wherein the upper key assembly comprises an upper lock feature configured to hold the upper key in the corresponding upper key slot, and wherein a retrieval tool is configured to jar the upper key assembly to shear the upper lock feature and release the upper key to retract from the upper key slot, and wherein retracting the upper key is configured to release the upper key assembly and the tubular opening prong from the primary flow tube for removal from the wellbore.

Statement 18. The subsurface safety valve system of statement 16 or statement 17, wherein the upper key assembly comprises an upper lip formed in a radially inner surface of the upper key assembly, wherein the upper lip is configured to interface with the retrieval tool.

Statement 19. The subsurface safety valve system of any of statements 16-18, wherein the secondary valve comprises a lower lock feature configured to hold the lower key in the corresponding lower key slot, and wherein a retrieval tool is configured to jar the secondary valve to shear the lower lock feature and release the lower key to retract from the lower key slot, and wherein retracting the lower key is configured to release the secondary valve from the valve housing.

Statement 20. A method for actuating a secondary subsurface safety valve, comprising: applying a threshold pressure to a control line disposed within a main body to drive a primary flow tube in an axially downhole direction with respect to the main body, wherein a retrievable lock mandrel is secured to the primary flow tube and an opening prong is secured to the retrievable lock

mandrel such that the opening prong moves in the axially downhole direction to contact and drive a secondary flapper to an open position in response to movement of the primary flow tube in the axially downhole direction, and wherein the secondary flapper is secured to a secondary valve seat that is secured to the main body in a position downhole from the primary flow tube; and releasing pressure to the control line, wherein a spring assembly biases the primary flow tube in an axially uphole direction in response to pressure in the control line falling below the threshold pressure, wherein the opening prong moves in the axially uphole direction to release the secondary flapper to move into a closed position in response to the primary flow tube moving in the axially uphole direction.

(45) For the sake of brevity, only certain ranges are explicitly disclosed herein. However, ranges from any lower limit may be combined with any upper limit to recite a range not explicitly recited, as well as, ranges from any lower limit may be combined with any other lower limit to recite a range not explicitly recited, in the same way, ranges from any upper limit may be combined with any other upper limit to recite a range not explicitly recited. Additionally, whenever a numerical range with a lower limit and an upper limit is disclosed, any number and any included range falling within the range are specifically disclosed. In particular, every range of values (of the form, “from about a to about b,” or, equivalently, “from approximately a to b,” or, equivalently, “from approximately a-b”) disclosed herein is to be understood to set forth every number and range encompassed within the broader range of values even if not explicitly recited. Thus, every point or individual value may serve as its own lower or upper limit combined with any other point or individual value or any other lower or upper limit, to recite a range not explicitly recited.

(46) The particular embodiments disclosed above are illustrative only. Although individual embodiments are discussed, all combinations of each embodiment are contemplated and covered by the disclosure. Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular illustrative embodiments disclosed above are non-exhaustive, and that other embodiments may be devised that are also within the scope of the present disclosure.

Claims

1. A subsurface safety valve system, comprising: a tubular valve housing disposable within a wellbore; a primary flow tube moveably secured within the valve housing between an upper position and a lower position; a secondary valve securable within the valve housing downhole from the primary flow tube and a primary valve, wherein the secondary valve is actuatable between an open position and a closed position; and a tubular opening prong securable to the primary flow tube, wherein the tubular opening prong is moveable axially downhole to drive the secondary valve into the open position in response to the primary flow tube moving to the lower position, and wherein the tubular opening prong is movable axially uphole to release the secondary valve to move into the closed position; and an upper key assembly having at least one upper key, wherein the at least one upper key is expandable into a corresponding upper key slot formed in the primary flow tube to secure the upper key assembly to the primary flow tube, wherein the upper key is retractable from the upper key slot to release the upper key assembly from the primary flow tube, and wherein the tubular opening prong is secured to a lower end of the upper key assembly.
2. The subsurface safety valve system of claim 1, wherein the tubular valve housing comprises a main body portion and an end sub secured to a downhole end of the main body portion, and wherein the secondary valve is engageable the end sub to secure the secondary valve to the end sub of the tubular valve housing.
3. The subsurface safety valve system of claim 2, wherein the primary valve is secured within the main body portion of the tubular valve housing.
4. The subsurface safety valve system of claim 1, wherein the primary valve comprises: a primary

valve seat secured within a main body portion of the valve housing; and a primary flapper secured to the primary valve seat, wherein the tubular opening prong is configured to hold the primary flapper in an open position.

5. The subsurface safety valve system of claim 1, further comprising a fluid control line and piston disposed within the tubular valve housing, wherein the piston is moveable between a first and second position in response to fluid pressure in the fluid control line, and wherein movement of the piston from the first position to the second position drives the primary flow tube from the upper position to the lower position.

6. The subsurface safety valve system of claim 5, further comprising a spring assembly disposed in the valve housing, wherein the spring assembly is configured to bias the piston from the second position toward the first position.

7. The subsurface safety valve system of claim 1, wherein the upper key assembly comprises an upper body portion and a lower body portion, wherein the upper body portion is axially moveable with respect to the lower body portion between an extended position and a collapsed position.

8. The subsurface safety valve system of claim 7, wherein the upper body portion comprises a shim extending from a downhole end of the upper body portion, wherein the shim contacts the at least one upper key in the collapsed position to bias the at least one upper key in a radially outward direction into the corresponding upper key slot formed in the primary flow tube.

9. The subsurface safety valve system of claim 8, further comprising an upper lock feature configured to engage and hold the shim in the collapsed position, wherein the upper lock feature comprises a shear pin, a shear screw, or some combination thereof.

10. The subsurface safety valve system of claim 7, wherein the upper body portion comprises a hollow mandrel having an upper lip formed in a radially inner surface of the upper body portion, wherein the upper lip is configured to interface with a retrieval tool, and wherein the retrieval tool is configured to jar the upper body portion, via the upper lip, to shear an upper lock feature holding the upper body portion in the collapsed position.

11. The subsurface safety valve system of claim 1, wherein the secondary valve comprises: a lower key assembly having at least one lower key, wherein the at least one lower key is expandable into a corresponding lower key slot formed in the valve housing to secure the secondary valve to the valve housing, and wherein the lower key is retractable from the lower key slot to release the secondary valve from the valve housing; a secondary valve seat connected to a downhole end of the lower key assembly; and a secondary flapper secured to the secondary valve seat, wherein the secondary flapper is actuatable between an open position and a closed position.

12. The subsurface safety valve system of claim 11, wherein the lower key assembly comprises a shifting portion and a seal portion, wherein the seal portion comprises a mandrel and a seal secured to an outer surface of the mandrel, wherein the seal is configured to interface with the valve housing, and wherein the shifting portion is moveable axially with respect to the seal portion between an extended position and a mated position.

13. The subsurface safety valve system of claim 12, wherein the shifting portion comprises a shim extending from a downhole end of the shifting portion, wherein the shim is configured to contact the at least one lower key in the mated position to bias the at least one lower key in a radially outward direction into the corresponding lower key slot formed in the valve housing.

14. The subsurface safety valve system of claim 12, wherein the shifting portion comprises a hollow mandrel having an inner lip formed at an uphole end of the shifting portion, wherein the inner lip is configured to interface with a retrieval tool with the tubular opening prong removed from the wellbore, and wherein the retrieval tool is configured to jar the shifting portion, via the inner lip, to shear a lower lock feature holding the shifting portion in the mated position.

15. A subsurface safety valve system, comprising: a tubular valve housing disposable within a wellbore, the valve housing having a lower key slot; a primary flow tube axially moveable within the valve housing between an upper position and a lower position, the primary flow tube having an

upper key slot formed in a radially inner surface of the primary flow tube; a secondary valve securable within the valve housing downhole from the primary flow tube and a primary valve, wherein the secondary valve comprises a lower key assembly having a lower key configured to expand into the corresponding lower key slot formed in the valve housing to secure the secondary valve to the valve housing; a secondary valve seat connected to a downhole end of the lower key assembly; a secondary flapper secured to the secondary valve seat, wherein the secondary flapper is actuatable between an open position and a closed position; an upper key assembly comprising an upper key, wherein the upper key is expandable into the upper key slot to secure the upper key assembly to the primary flow tube; and a tubular opening prong secured to a downhole end of the upper key assembly, wherein the opening prong is moveable axially downhole to drive the secondary flapper into the open position in response to the primary flow tube moving to the lower position, and wherein the opening prong is moveable axially uphole to release the secondary flapper to move into the closed position.

16. The subsurface safety valve system of claim 15, wherein the upper key assembly comprises an upper lock feature configured to hold the upper key in the corresponding upper key slot, and wherein a retrieval tool is configured to jar the upper key assembly to shear the upper lock feature and release the upper key to retract from the upper key slot, and wherein retracting the upper key is configured to release the upper key assembly and the tubular opening prong from the primary flow tube for removal from the wellbore.

17. The subsurface safety valve system of claim 16, wherein the upper key assembly comprises an upper lip formed in a radially inner surface of the upper key assembly, wherein the upper lip is configured to interface with the retrieval tool.

18. The subsurface safety valve system of claim 15, wherein the secondary valve comprises a lower lock feature configured to hold the lower key in the corresponding lower key slot, and wherein a retrieval tool is configured to jar the secondary valve to shear the lower lock feature and release the lower key to retract from the lower key slot, and wherein retracting the lower key is configured to release the secondary valve from the valve housing.

19. A method for actuating a secondary subsurface safety valve, comprising: applying a threshold pressure to a control line disposed within a main body to drive a primary flow tube in an axially downhole direction with respect to the main body, wherein a retrievable lock mandrel is secured to the primary flow tube and an opening prong is secured to the retrievable lock mandrel such that the opening prong moves in the axially downhole direction to contact and drive a secondary flapper to an open position in response to movement of the primary flow tube in the axially downhole direction, and wherein the secondary flapper is secured to a secondary valve seat that is secured to the main body in a position downhole from the primary flow tube; and releasing pressure to the control line, wherein a spring assembly biases the primary flow tube in an axially uphole direction in response to pressure in the control line falling below the threshold pressure, wherein the opening prong moves in the axially uphole direction to release the secondary flapper to move into a closed position in response to the primary flow tube moving in the axially uphole direction.
